

Homework (Carolina Reaper)

- Q1.** What is the absolute value of x on a number line?
- (A) x itself
 - (B) the distance of x from zero
 - (C) the negative of x
 - (D) the square of x
 - (E) the cube root of x
- Q2.** A music producer wants to mix two sound waves with frequencies of -25 Hz and 30 Hz. What is the absolute value of the difference between these frequencies?
- (A) 5 Hz
 - (B) 35 Hz
 - (C) 45 Hz
 - (D) 55 Hz
 - (E) 65 Hz
- Q3.** An artist is creating a painting on a canvas that is 12 feet wide and -8 feet long. What is the absolute value of the length?
- (A) 4 feet
 - (B) 6 feet
 - (C) 8 feet
 - (D) 10 feet
 - (E) 12 feet
- Q4.** Evaluate the expression $|2 \cdot (-3)|$.
- (A) 6
 - (B) -6
 - (C) 4
 - (D) -4
 - (E) 0
- Q5.** A musician is playing a rhythm pattern with a tempo of -120 beats per minute (bpm). What is the absolute value of this tempo?
- (A) 100 bpm
 - (B) 110 bpm
 - (C) 120 bpm
 - (D) 130 bpm
 - (E) 140 bpm
- Q6.** Solve for x in the equation $|x| = 9$.
- (A) $x = 9$
 - (B) $x = -9$
 - (C) $x = 0$
 - (D) $x = 9$ or $x = -9$
 - (E) $x = 3$ or $x = -3$
- Q7.** A sound engineer is adjusting the decibel level of a speaker, which is currently at -20 decibels. What is the absolute value of this decibel level?
- (A) 10 decibels
 - (B) 15 decibels
 - (C) 20 decibels
 - (D) 25 decibels
 - (E) 30 decibels
- Q8.** A painter is mixing two colors with pigment values of 15 and -20 . What is the absolute value of the difference between these pigment values?
- (A) 5
 - (B) 15
 - (C) 25
 - (D) 30
 - (E) 35

Open Ended Questions (FRQ)

- Q9.** A musician is playing a song with a frequency of -18 Hz. If the musician wants to increase the frequency by 12 Hz, what will be the new frequency? Show your work and explain your answer in the space provided below.
- Q10.** An artist is creating a sculpture with a height of -25 inches and a width of 30 inches. What is the absolute value of the difference between the height and the width? Show your work and explain your answer in the space provided below.

Chef's Tips

- Tip 1: Ensure students understand the concept of absolute value as distance from zero.
- Tip 2: Encourage students to use real-world examples, such as music and art, to illustrate absolute value.
- Tip 3: Remind students to show their work and explain their answers clearly.